

Probability

ANSWER KEY



Basic probability

- 1 Two fair six-sided dice are rolled. Find the probability their sum is 8.

Total outcomes: $n(S) = 36$. Favorable outcomes (sum = 8): (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) — that's 5.

$$\text{So } P(\text{sum} = 8) = \frac{5}{36}.$$

Complement rule

- 2 If $P(E) = 0.35$, find $P(E')$.

By Definition 6.2, $P(E') = 1 - P(E) = 1 - 0.35 = 0.65$.

Permutations and combinations

- 3 How many ways can a committee of 4 be chosen from 10 people?

Order doesn't matter, so use combinations: $\binom{10}{4} = \frac{10!}{4! 6!} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = 210$.

- 4 How many ways can 2 officers (president, vice president — order matters) be chosen from 6 candidates?

Order matters, so use permutations: $P(6, 2) = \frac{6!}{4!} = 6 \cdot 5 = 30$.

- 5 Explain, using Theorem 6.1, how $\binom{6}{2}$ relates to $P(6, 2)$ from the previous question.

Theorem 6.1 says $\binom{6}{2} = \frac{P(6, 2)}{2!} = \frac{30}{2} = 15$ — combinations undercount permutations by exactly the 2! orderings of each chosen pair.

Independent events and the addition rule

- 6 Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

By Definition 6.4, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.4)(0.5) = 0.2$.

- 7 A card is drawn from a standard 52-card deck. Find $P(\text{Queen or Spade})$.

$P(\text{Queen}) = \frac{4}{52}$, $P(\text{Spade}) = \frac{13}{52}$, and $P(\text{Queen and Spade}) = \frac{1}{52}$ (the Queen of Spades).

By the addition rule: $P(\text{Queen or Spade}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$.

Multiple choice — easy

- 8 A bag contains 9 yellow marbles, 5 red marbles, and 7 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{1}{3}$

b. $\frac{3}{7}$ ✓

c. $\frac{10}{21}$

d. $\frac{5}{21}$

Total marbles: $n(S) = 21$. $P(\text{yellow}) = \frac{9}{21} = \frac{3}{7}$.

- 9 A bag contains 3 green marbles, 7 yellow marbles, and 7 red marbles. A marble is drawn at random. Find $P(\text{green})$.

a. $\frac{7}{17}$

b. $\frac{5}{17}$

c. $\frac{4}{17}$

d. $\frac{3}{17}$ ✓

Total marbles: $n(S) = 17$. $P(\text{green}) = \frac{3}{17}$.

- 10 A bag contains 9 yellow marbles, 9 red marbles, and 6 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{5}{12}$

b. $\frac{11}{24}$

c. $\frac{1}{4}$

d. $\frac{3}{8}$ ✓

Total marbles: $n(S) = 24$. $P(\text{yellow}) = \frac{9}{24} = \frac{3}{8}$.

- 11 A bag contains 5 yellow marbles, 3 red marbles, and 7 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{7}{15}$

b. $\frac{2}{5}$

c. $\frac{1}{3}$ ✓

d. $\frac{1}{5}$

Total marbles: $n(S) = 15$. $P(\text{yellow}) = \frac{5}{15} = \frac{1}{3}$.

- 12 A bag contains 7 blue marbles, 7 green marbles, and 3 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{9}{17}$

b. $\frac{8}{17}$

c. $\frac{3}{17}$

d. $\frac{7}{17}$ ✓

Total marbles: $n(S) = 17$. $P(\text{blue}) = \frac{7}{17}$.

- 13 A bag contains 5 yellow marbles, 4 red marbles, and 4 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{5}{13}$ ✓

b. $\frac{4}{13}$

c. $\frac{6}{13}$

d. $\frac{7}{13}$

Total marbles: $n(S) = 13$. $P(\text{yellow}) = \frac{5}{13}$.

- 14 A bag contains 7 blue marbles, 7 green marbles, and 5 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{7}{19}$ ✓

b. $\frac{8}{19}$

c. $\frac{9}{19}$

d. $\frac{5}{19}$

Total marbles: $n(S) = 19$. $P(\text{blue}) = \frac{7}{19}$.

- 15 A bag contains 2 blue marbles, 4 green marbles, and 4 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{2}{5}$

b. $\frac{1}{5}$ ✓

c. $\frac{3}{10}$

d. $\frac{1}{2}$

Total marbles: $n(S) = 10$. $P(\text{blue}) = \frac{2}{10} = \frac{1}{5}$.

- 16 A bag contains 4 blue marbles, 5 green marbles, and 5 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{2}{7}$ ✓

b. $\frac{1}{2}$

c. $\frac{3}{7}$

d. $\frac{5}{14}$

Total marbles: $n(S) = 14$. $P(\text{blue}) = \frac{4}{14} = \frac{2}{7}$.

- 17 A bag contains 3 yellow marbles, 3 red marbles, and 2 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

- a. $\frac{1}{2}$
- b. $\frac{1}{4}$
- c. $\frac{5}{8}$
- d. $\frac{3}{8}$ ✓

Total marbles: $n(S) = 8$. $P(\text{yellow}) = \frac{3}{8}$.

- 18 A bag contains 9 blue marbles, 2 green marbles, and 4 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

- a. $\frac{4}{15}$
- b. $\frac{2}{15}$
- c. $\frac{2}{3}$
- d. $\frac{9}{15}$ ✓

Total marbles: $n(S) = 15$. $P(\text{blue}) = \frac{9}{15} = \frac{3}{5}$.

- 19 A bag contains 5 green marbles, 3 yellow marbles, and 3 red marbles. A marble is drawn at random. Find $P(\text{green})$.

- a. $\frac{6}{11}$
- b. $\frac{3}{11}$
- c. $\frac{7}{11}$
- d. $\frac{5}{11}$ ✓

Total marbles: $n(S) = 11$. $P(\text{green}) = \frac{5}{11}$.

- 20 A bag contains 3 red marbles, 3 blue marbles, and 4 green marbles. A marble is drawn at random. Find $P(\text{red})$.

a. $\frac{3}{10}$ ✓

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $\frac{2}{5}$

Total marbles: $n(S) = 10$. $P(\text{red}) = \frac{3}{10}$.

- 21 A bag contains 6 red marbles, 8 blue marbles, and 9 green marbles. A marble is drawn at random. Find $P(\text{red})$.

a. $\frac{6}{23}$ ✓

b. $\frac{7}{23}$

c. $\frac{9}{23}$

d. $\frac{8}{23}$

Total marbles: $n(S) = 23$. $P(\text{red}) = \frac{6}{23}$.

- 22 A bag contains 5 green marbles, 9 yellow marbles, and 3 red marbles. A marble is drawn at random. Find $P(\text{green})$.

a. $\frac{3}{17}$

b. $\frac{9}{17}$

c. $\frac{5}{17}$ ✓

d. $\frac{6}{17}$

Total marbles: $n(S) = 17$. $P(\text{green}) = \frac{5}{17}$.

Multiple choice — medium

23 How many ways can a committee of 6 be chosen from 7 people?

- a. 1
- b. 3
- c. 11
- d. 7 ✓

Order doesn't matter, so use combinations: $\binom{7}{6} = 7$.

24 How many ways can a committee of 5 be chosen from 6 people?

- a. 10
- b. 1
- c. 2
- d. 6 ✓

Order doesn't matter, so use combinations: $\binom{6}{5} = 6$.

25 How many ways can a committee of 5 be chosen from 16 people?

- a. 4368 ✓
- b. 8008
- c. 4364
- d. 4372

Order doesn't matter, so use combinations: $\binom{16}{5} = 4368$.

26 How many ways can a committee of 4 be chosen from 11 people?

- a. 326
- b. 334
- c. 462
- d. 330 ✓

Order doesn't matter, so use combinations: $\binom{11}{4} = 330$.

27 How many ways can a committee of 3 be chosen from 14 people?

a. 364 ✓

b. 368

c. 1001

d. 360

Order doesn't matter, so use combinations: $\binom{14}{3} = 364$.

28 How many ways can a committee of 2 be chosen from 20 people?

a. 194

b. 190 ✓

c. 186

d. 1140

Order doesn't matter, so use combinations: $\binom{20}{2} = 190$.

29 How many ways can a committee of 3 be chosen from 10 people?

a. 120 ✓

b. 116

c. 124

d. 210

Order doesn't matter, so use combinations: $\binom{10}{3} = 120$.

30 How many ways can a committee of 2 be chosen from 19 people?

- a. 175
- b. 167
- c. 969
- d. 171 ✓

Order doesn't matter, so use combinations: $\binom{19}{2} = 171$.

31 How many ways can a committee of 4 be chosen from 9 people?

- a. 126 ✓
- b. 122
- c. 130
- d. 132

Order doesn't matter, so use combinations: $\binom{9}{4} = 126$.

32 How many ways can a committee of 3 be chosen from 15 people?

- a. 1365
- b. 459
- c. 455 ✓
- d. 451

Order doesn't matter, so use combinations: $\binom{15}{3} = 455$.

33 How many ways can a committee of 5 be chosen from 17 people?

- a. 6192
- b. 12376
- c. 6188 ✓
- d. 6184

Order doesn't matter, so use combinations: $\binom{17}{5} = 6188$.

34 How many ways can a committee of 2 be chosen from 18 people?

a. 153 ✓

b. 149

c. 157

d. 816

Order doesn't matter, so use combinations: $\binom{18}{2} = 153$.

35 How many ways can a committee of 6 be chosen from 18 people?

a. 18564 ✓

b. 31824

c. 18560

d. 18568

Order doesn't matter, so use combinations: $\binom{18}{6} = 18564$.

36 How many ways can a committee of 2 be chosen from 14 people?

a. 87

b. 91 ✓

c. 364

d. 95

Order doesn't matter, so use combinations: $\binom{14}{2} = 91$.

37 How many ways can a committee of 5 be chosen from 11 people?

- a. 458
- b. 462 ✓
- c. 468
- d. 466

Order doesn't matter, so use combinations: $\binom{11}{5} = 462$.

Multiple choice — hard

38 How many ways can 2 distinct positions be filled from 12 candidates, where order matters?

- a. 66
- b. 138
- c. 132 ✓
- d. 126

Order matters, so use permutations: $P(12, 2) = \frac{12!}{10!} = 12 \times 11 = 132$.

39 How many ways can 4 distinct positions be filled from 9 candidates, where order matters?

- a. 3024 ✓
- b. 126
- c. 3018
- d. 3030

Order matters, so use permutations: $P(9, 4) = \frac{9!}{5!} = 9 \times 8 \times 7 \times 6 = 3024$.

40 How many ways can 4 distinct positions be filled from 14 candidates, where order matters?

a. 24018

b. 1001

c. 24030

d. 24024 ✓

Order matters, so use permutations: $P(14, 4) = \frac{14!}{10!} = 14 \times 13 \times 12 \times 11 = 24024$.

41 How many ways can 2 distinct positions be filled from 11 candidates, where order matters?

a. 110 ✓

b. 55

c. 116

d. 104

Order matters, so use permutations: $P(11, 2) = \frac{11!}{9!} = 11 \times 10 = 110$.

42 How many ways can 4 distinct positions be filled from 15 candidates, where order matters?

a. 32760 ✓

b. 32766

c. 32754

d. 1365

Order matters, so use permutations: $P(15, 4) = \frac{15!}{11!} = 15 \times 14 \times 13 \times 12 = 32760$.

43 Events A and B are independent, with $P(A) = 0.1$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

a. $\frac{1}{2}$

b. $\frac{9}{100}$

c. $\frac{19}{100}$

d. $\frac{1}{25}$ ✓

By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.1)(0.4) = \frac{1}{25}$.

44 Events A and B are independent, with $P(A) = 0.2$ and $P(B) = 0.3$. Find $P(A \text{ and } B)$.

a. $\frac{1}{2}$

b. $\frac{1}{100}$

c. $\frac{11}{100}$

d. $\frac{3}{50}$ ✓

By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.2)(0.3) = \frac{3}{50}$.

45 Events A and B are independent, with $P(A) = 0.5$ and $P(B) = 0.1$. Find $P(A \text{ and } B)$.

a. $\frac{1}{5}$

b. $\frac{3}{5}$

c. $\frac{1}{10}$

d. $\frac{1}{20}$ ✓

By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.5)(0.1) = \frac{1}{20}$.

46 Events A and B are independent, with $P(A) = 0.2$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

a. $\frac{13}{100}$

b. $\frac{3}{100}$

c. $\frac{2}{25}$ ✓

d. $\frac{3}{5}$

By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.2)(0.4) = \frac{2}{25}$.

47 Events A and B are independent, with $P(A) = 0.6$ and $P(B) = 0.3$. Find $P(A \text{ and } B)$.

- a. $\frac{23}{100}$
- b. $\frac{13}{100}$
- c. $\frac{9}{50}$ ✓
- d. $\frac{9}{10}$

By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.6)(0.3) = \frac{9}{50}$.

48 Out of 38 equally likely tiles, 5 are red, 8 are numbered, and 3 are both. One item is chosen at random. Find $P(\text{red or numbered})$.

- a. $\frac{5}{19}$ ✓
- b. $\frac{6}{19}$
- c. $\frac{13}{38}$
- d. $\frac{4}{19}$

By the addition rule: $P(\text{red or numbered}) = \frac{5}{38} + \frac{8}{38} - \frac{3}{38} = \frac{10}{38} = \frac{5}{19}$.

49 Out of 27 equally likely boxes, 7 are labeled, 8 are sealed, and 7 are both. One item is chosen at random. Find $P(\text{labeled or sealed})$.

- a. $\frac{2}{9}$
- b. $\frac{10}{27}$
- c. $\frac{8}{27}$ ✓
- d. $\frac{5}{9}$

By the addition rule: $P(\text{labeled or sealed}) = \frac{7}{27} + \frac{8}{27} - \frac{7}{27} = \frac{8}{27}$.

- 50 Out of 24 equally likely boxes, 4 are labeled, 8 are sealed, and 3 are both. One item is chosen at random. Find $P(\text{labeled or sealed})$.

a. $\frac{11}{24}$

b. $\frac{3}{8}$ ✓

c. $\frac{7}{24}$

d. $\frac{1}{2}$

By the addition rule: $P(\text{labeled or sealed}) = \frac{4}{24} + \frac{8}{24} - \frac{3}{24} = \frac{9}{24} = \frac{3}{8}$.

- 51 Out of 26 equally likely tickets, 9 are gold, 12 are striped, and 8 are both. One item is chosen at random. Find $P(\text{gold or striped})$.

a. $\frac{21}{26}$

b. $\frac{15}{26}$

c. $\frac{1}{2}$ ✓

d. $\frac{11}{26}$

By the addition rule: $P(\text{gold or striped}) = \frac{9}{26} + \frac{12}{26} - \frac{8}{26} = \frac{13}{26} = \frac{1}{2}$.

- 52 Out of 37 equally likely numbers, 4 are even, 10 are prime, and 2 are both. One item is chosen at random. Find $P(\text{even or prime})$.

a. $\frac{14}{37}$

b. $\frac{10}{37}$

c. $\frac{12}{37}$ ✓

d. $\frac{15}{37}$

By the addition rule: $P(\text{even or prime}) = \frac{4}{37} + \frac{10}{37} - \frac{2}{37} = \frac{12}{37}$.